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examples of a discrete LTI system response solution

[Example] The differential equation for causal discrete LTI systems is

$y[k]-3y[k-1]+2y[k-2]=x[k],\ k\geq 0$

Drive signal $x[k]=3^k u[k]$, initial state $y[-1]=3$, $y[-2]=1$, try to find：

(1) The zero input response of the system $y_{zi}[k]$

(2) The unit impulse response of the system $h[k]$

(3) The zero-state response of the system $y_{zs}[k]$

(4) The system is fully responsive $y[k]$

(5) Determine whether the system is stable.

examples of a discrete LTI system response solution

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(1) The zero input response of the system $y_{zi}[k]$

Eigen-equations $r^2-3r+2=0$

The feature root is $r_1=1\ ,\quad r_2=2$

$$\Rightarrow \quad y_{zi}[k]=A+B2^k,\quad k\geq 0$$

Substituting into the initial state,

$$y[-1]=A+B/2=3$$

$$y[-2]=A+B/4=1$$

$$\Rightarrow \quad A=-1,\ B=8$$

$$y_{zi}[k]=-1+8\cdot 2^k,\quad k\geq 0$$

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Drive signal $x[k]=3^k u[k]$, initial state $y[-1]=3$, $y[-2]=1$, try to find:

(2) Unit impulse response $h[k]$

$$h[k]-3h[k-1]+2h[k-2]=\delta[k]$$

The unit impulse response is in the form of $h[k]$.

$$h[k]=Cu[k]+D2^k u[k]$$

The equivalent initial condition is

$$h[0]=\delta[0]+3h[-1]-2h[-2]=1$$

$$h[1]=\delta[1]+3h[0]-2h[-1]=3$$

$$h[0]=C+D=1$$

$$h[1]=C+2D=3 \quad \Rightarrow \quad C=-1 \qquad D=2$$

$$h[k]=-u[k]+2\cdot 2^k u[k]$$

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(3) The zero-state response of the system $y_{zs}[k]$

Utilize convolution and the zero-state response of the system to be found $y_{zs}[k]$

$$y_{zs}[k]=x[k]*h[k]=3^k u[k]*(2\cdot 2^k-1)u[k]$$

$$=(\frac{9}{2}3^k-4\cdot 2^k+\frac{1}{2})u[k]$$

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[Example] The differential equation for causal discrete LTI systems is

$y[k]-3y[k-1]+2y[k-2]=x[k],\ k\geq 0$

Drive signal $x[k]=3^k u[k]$, initial state $y[-1]=3$, $y[-2]=1$, try to find：

(4) The system is fully responsive $y[k]$

$$y[k]=y_{\mathrm{zi}}[k]+y_{zs}[k]$$

$$=\frac{9}{2}\cdot 3^k+4\cdot 2^k-\frac{1}{2},\quad k\geq 0$$

examples of a discrete LTI system response solution

[Example] The differential equation for causal discrete LTI systems is

$y[k]-3y[k-1]+2y[k-2]=x[k],\ k\geq 0$

Drive signal $x[k]=3^k u[k]$, initial state $y[-1]=3$, $y[-2]=1$, try to find:

(5) Determine whether the system is stable

The unit impulse response of the system is $h[k]=(2\cdot 2^k-1)u[k]$

$$\sum_{k=-\infty}^{\infty}|h[k]|=\sum_{k=0}^{\infty}(2\cdot 2^k-1)=\infty$$

The discrete LTI system is an unstable system.

examples of a discrete LTI system response solution

[Example] The excitation signal is changed from $x[k]=3^k u[k]$ to $0.5\cdot 3^{k-1}u[k-1]$ recalculated to the system:

Zero input response $y_{zi}[k]$、Zero input response $y_{zs}[k]$ and fully responsive $y[k]$

Solution:

Since the initial state of the system has not changed, the zero input response of the system does not change, that is

$$y_{zi}[k]=-1+8\cdot 2^k,\quad k\geq 0$$

The excitation signal is $0.5\cdot 3^{k-1}u[k-1]=0.5x[k-1]$

So using the linear and non-time-varying characteristics of the system,

The zero-state response of the available system is

$$0.5y_{zs}[k-1]=(\frac{9}{4}\cdot 3^{k-1}-2\cdot 2^{k-1}+\frac{1}{4})u[k-1]$$

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[Example] The excitation signal is changed from $x[k]=3^k u[k]$ to $0.5\cdot 3^{k-1}u[k-1]$ recalculated to the system:

Zero input response $y_{zi}[k]$、Zero input response $y_{zs}[k]$ and fully responsive $y[k]$

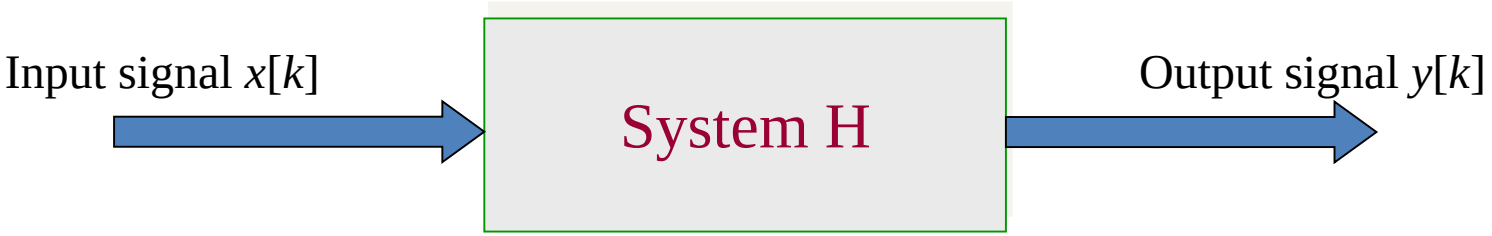
The zero input response of the system $y_{zi}[k]=-1+8\cdot 2^k \quad ,k\geq 0$

The zero-state response of the system $y_{zs}[k]=(\frac{9}{4}\cdot 3^{k-1}-2\cdot 2^{k-1}+\frac{1}{4})u[k-1]$

The system is fully responsive $y[k]=y_{zi}[k]+y_{zs}[k]$

$$=(-1+8\cdot 2^k)u[k]+(\frac{9}{4}\cdot 3^{k-1}-2\cdot 2^{k-1}+\frac{1}{4})u[k-1]$$

Summary of time domain analysis of discrete LTI systems



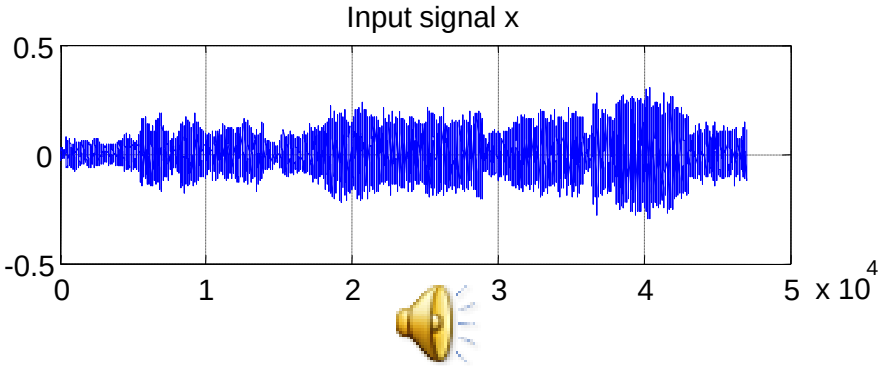
(a) The input $\boldsymbol{x}$ and system H are known, and the output $\boldsymbol{y}$ is solved, which is called a positive problem.

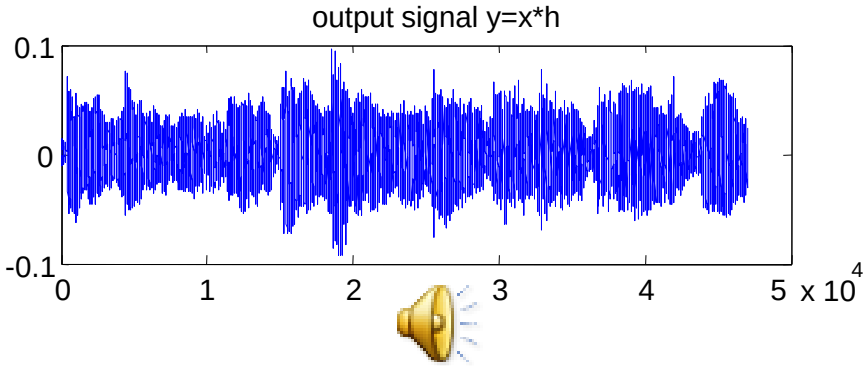
(b) Known output $\boldsymbol{y}$ and system H, solve input $\boldsymbol{x}$; Or the known output $\boldsymbol{y}$ and input $\boldsymbol{x}$, to solve the system H; called the inverse problem.

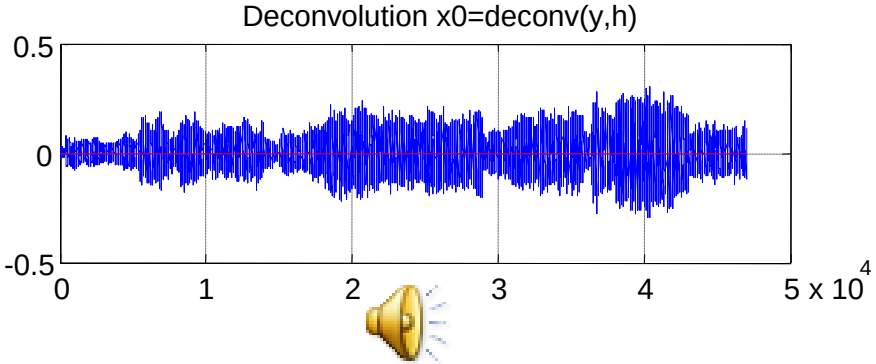
(c) Known output $\boldsymbol{y}$ and part of the system H, solve input $\boldsymbol{x}$; Or known output $\boldsymbol{y}$ and partial input $\boldsymbol{x}$, solve system H, called the gray box problem.

(d) Known output y, unknown input $\boldsymbol{x}$, and system H, known as black box problem.

Summary of time domain analysis of discrete LTI systems







h[k]=[1/8,1/8,1/8,1/8, 1/8, 1/8, 1/8, 1/8]

Summary of time domain analysis of discrete LTI systems

summary:

1. The time domain analysis of the system gives a time domain description of the discrete LTI system.

2. The time domain analysis of the system reveals the mechanism of action of the signal and the system.

3. The time domain analysis of the system is based on the time domain analysis of discrete signals

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!